Bile Acid Report Comparison and Verification Audit

Inputs reviewed: Bile_Acid_LMM_Analysis_Report.pdf, Bile_Acid_Analysis_Report.pdf, Bile_Acid_LMM_Results.xlsx, IRMA23930_MS-BA_Results.csv, and the local R scripts. Generated May 20, 2026.

Executive Verdict

Most reliable answer for the main post-meal question: the audited finite-sample LMM results in *Bile_Acid_LMM_Analysis_Report.pdf* are the stronger result set. The original XLSX p-values are reproducible, but they are asymptotic z-test p-values; the finite-sample Satterthwaite/Kenward-Roger correction is more appropriate for 32 subjects.

Use the CSV/R PDF with caution. Its individual bile-acid pairwise model estimates are reproducible, but the PDF is not a clean final statistical report: it reports raw p-value significance rather than FDR q-values, answers a different contrast structure, contains aggregate-class missing-data problems, and includes narrative claims that contradict its own tables.

Not an apples-to-apples disagreement: the LMM report tests timepoint 23 against the average of the fasting states (2, 18, 26), while the CSV/R report tests three separate pairwise contrasts: 23 vs 2, 23 vs 18, and 23 vs 26.

Data Lineage Checks

Check	Verification result	Implication
Raw CSV vs XLSX raw sheet	Same 203 x 19 assay table; maximum numeric difference approximately 1.4e-14.	Both reports start from the same raw data.
XLSX batch-adjusted sheet	Exactly matches LTR mean-based plate factors (max difference approximately 9.1e-13).	The XLSX method is mean-based, not median-based.
CSV/R script batch adjustment	Uses LTR median-based plate factors.	This is defensible but different from the XLSX pipeline; results are not expected to be identical.
Timepoints analyzed	Both exclude timepoint 10 and use 2, 18, 23, and 26.	The core longitudinal subset is aligned.

Major Method Differences

Area	LMM/XLSX audit report	CSV/R analysis report
Batch correction	LTR mean-based adjustment in the workbook.	LTR median-based adjustment in run_analysis.R.
Transform	log10(concentration).	Natural log of concentration; adds 1 if any zero is present.
Primary endpoint	23 vs average fasting state (2, 18, 26).	Separate 23 vs 2, 23 vs 18, and 23 vs 26 contrasts.
Multiple testing	BH FDR over 15 bile acids; PDF explicitly audits asymptotic vs finite-sample p-values.	Existing PDF shows raw p-value stars. Current R script adds FDR, but the PDF in the folder is stale.
Degrees of freedom	Audit identifies finite-sample correction as best practice.	Text says Satterthwaite; current emmeans default in this environment is Kenward-Roger unless lmer.df is set explicitly.
Aggregate classes	Not the focus of the workbook LMM results.	Aggregate classes have a missing-data bug: all-missing component rows become zero via rowSums(..., na.rm=TRUE).

Verified Key Results

For the LMM/XLSX endpoint (23 vs average fasting), the workbook p-values match R only when using asymptotic degrees of freedom. Recomputing the same model with finite-sample correction changes the borderline calls.

Compound	Fold change 23 / fasting avg	Satterthwaite p	Satterthwaite FDR q	Sig
TCDCA	1.58	<0.001	0.003	**
GLCA	1.94	<0.001	<0.001	***
TLCA	1.57	0.003	0.010	*
GCA	1.31	0.018	0.045	*
TCA	1.42	0.004	0.013	*
TDCA	1.56	0.001	0.007	**

Borderline And Changed Calls

Compound	XLSX/asymptotic FDR q	XLSX sig	Finite-sample FDR q	Finite-sample sig	Interpretation
GCDCA	0.053	ns	0.058	ns	Borderline, remains non-significant after FDR.
LCA	0.045	*	0.050	ns	False positive in asymptotic XLSX call; finite-sample q is just above 0.05.
TLCA	0.008	**	0.010	*	Still significant, but one star weaker.
TCA	0.010	**	0.013	*	Still significant, but one star weaker.

CSV/R Pairwise Results After FDR

Contrast	FDR-significant individual bile acids	Important correction to PDF interpretation
23 vs baseline 2	GCDCA, TCDCA, GLCA, TLCA, GCA, TCA, TDCA	Consistent strong post-meal signal in conjugated acids.
23 vs NS4 fasting 18	TCDCA, LCA, GLCA, TLCA, TCA, TDCA	GCA is raw-p significant in the PDF but not FDR significant.
23 vs RTDS fasting 26	LCA, GLCA	Only LCA and GLCA survive FDR.

Aggregate-Class Issue In The CSV/R Report

The aggregate-class section is the least reliable part of the CSV/R PDF. The script calculates aggregate pools with rowSums(..., na.rm = TRUE). When every component in an aggregate is missing, R returns 0, not NA. The script then includes that row and models log(value + 1). This created one extra aggregate observation (N=127 instead of N=126) and materially changed p-values.

Aggregate	PDF-style N	PDF-style q 23 vs 2	PDF-style sig	Corrected N	Corrected q 23 vs 2	Corrected sig	Corrected q 23 vs 18	Corrected sig
Total_BA	127	0.207	ns	126	0.007	**	0.777	ns
Glycine_Conjugated	127	0.207	ns	126	0.007	**	0.224	ns
Taurine_Conjugated	127	0.001	**	126	<0.001	***	0.026	*
Unconjugated	127	0.871	ns	126	0.918	ns	0.521	ns

Internal Narrative Inconsistencies In The CSV/R PDF

Location	What the table says	What the narrative says	Why it matters
Glycine-conjugated aggregate	23 vs baseline p=0.0556, non-significant in the PDF table.	Claims a highly significant spike compared to all fasting points.	Conclusion overstates the reported table result.
Taurine-conjugated aggregate	23 vs RTDS fasting p=0.0985, non-significant.	Claims significance relative to all fasting points.	The RTDS comparison does not support that statement.
Unconjugated aggregate	All three contrasts are non-significant in the table.	Claims p<0.001 vs baseline and fasting 18.	This is a direct contradiction and should not be used.

Recommended Final Position

Question	Recommended answer
Which report is more likely correct?	Use the LMM audit report's finite-sample composite results as the primary answer for post-meal divergence.
Can the CSV/R report be used?	Use only as a pairwise exploratory analysis after applying FDR q-values and regenerating the PDF from the current script.
What should be fixed before publication?	Choose one TR normalization rule, set lmer.df explicitly, report FDR q-values, fix aggregate missingness, and rewrite narrative text from the tables.
Biological conclusion that survives both approaches	The strongest and most consistent postprandial signals are concentrated in conjugated bile acids, especially TCDCA, GLCA, TLCA, GCA, TCA, and TDCA.

Audit artifacts generated in comparison_work: lmm_composite_check.csv, r_pairwise_report_style_check.csv, and r_pairwise_corrected_missing_check.csv.